Selecting Reliable LLM Reasoning
with Answer-Group Preferences

A compact CPU selector for four fixed mathematical solutions

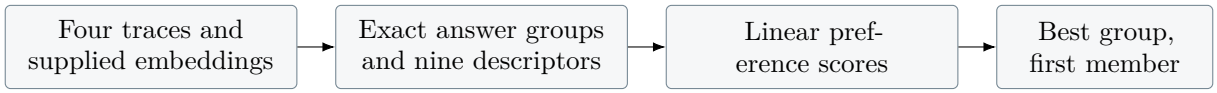
Abstract. This project studies answer selection rather than solution generation. Given four supplied traces and fixed embeddings, the selector must choose a candidate whose final numeric answer is correct. The final method pools numerically equivalent answers, combines their 1024-dimensional mean embeddings with nine group/answer descriptors, and fits a linear Bradley–Terry preference model. Matched training comparisons show that pooling alone hurts: it answers 1596/2114 out-of-fold questions correctly, versus 1682 for candidate-level preferences. Adding the descriptor block raises this to 1713, with 299/355 on informative validation questions. The frozen checkpoint obtains **387/467 (82.8694%)** on the official test bank and contains only **26,189 bytes** of inference assets. Feature ablations and nested evaluation of fallback rules support a conservative final design. In particular, a threshold rule apparently gaining 15 questions when selected on all OOF predictions loses two under nested evaluation. The principal findings concern representation, fair comparisons and evaluation reliability, rather than model size or a claim of verified mathematical reasoning.

1 Question and scope

The supplied solutions can be fluent, internally consistent, and still answer the wrong question. The course benchmark grades the selected *final number*, including equivalent fractions and decimals; it does not grade every reasoning step. We therefore distinguish selecting a correct answer from certifying a correct derivation. GSM8K provides the underlying word problems [2]; the candidate bank and representations are those supplied by the course [1].

The main research question is:

> Does exact numeric-answer grouping, together with a small block of group and answer descriptors, improve a fixed linear preference ranker under CPU limits? Which parts of the apparent improvement survive fair controls?



The study has three useful outcomes. First, it separates the benefit of the descriptor block from the unsuccessful pooling-only change. Second, it exposes how easy validation questions and reuse of OOF predictions can inflate impressions of reliability. Third, it retains a small, reproducible model after more elaborate alternatives fail to provide stable evidence of improvement. No external training data, additional pretrained weights, new solution text or model API is used by the selector.

2 Data, labels and evaluation protocol

Each question supplies four candidate texts and a float32 embedding array of shape 4×1024. The official parser maps each final answer to a canonical rational value, or returns invalid. For training candidate i in question q, the target is

$$y_{qi} = \mathbf{1}\{a_{qi} \neq \varnothing \ \wedge \ a_{qi} = a_q^{\text{ref}}\}.$$

Several candidates may be positive. Labels are aligned by both identifier and question text; identifiers are never predictive features.

Table 1: The informative subset has at least two distinct valid numeric answers.

	Train	Validation	Test
All supplied questions	4755	798	467
Four candidates all correct	2242	379	–
After removing all-correct questions	2513	419	–
Informative answer disagreement	2114	355	467
Non-all-correct, one valid answer group	399	64	0

Training support. Final weights and all normalization statistics use only the 2114 informative training questions. All-correct questions provide no positive-versus-negative ranking pairs. The further 399 questions contain one valid answer group and do not teach a preference between valid answers. The official files are not edited: filtering is an in-memory mask. Earlier MLP experiments already excluded all-correct questions from their loss, but used different standardization/support settings; this is a confound in historical cross-family comparisons.

Validation denominators. We report the complete 798-question bank, the 419 non-all-correct questions when useful, and primarily the 355 informative questions. Validation contains 443 one-valid-group questions; choosing that group answers all of them correctly. The final model's 742/798 thus decomposes as $443 + 299$. Full validation accuracy is not directly comparable to test, where all 467 questions require choosing between different valid answers.

OOF construction. Three-fold assignments are generated on all 4755 training rows, before informative filtering. With NumPy `default_rng` seed 20261028, rows are stratified by disagreement and number of positive candidates, shuffled within strata and assigned round-robin with a carried offset. Exact duplicate question texts share the first occurrence's fold. Each fit and its statistics use informative questions outside its held-out fold. The informative held-out sizes are 704, 705 and 705. Generating folds only on the 2114-question subset would change the random sequence and the experiment.

Development boundary. Validation was reused for development and is not a fresh holdout. Public test references were available for official scoring, and several frozen historical methods were scored during the project. They did not enter weight fitting, feature construction or threshold estimation. The reported test result is therefore a public-bank result, not a claim of a single-use blinded evaluation. No test error cases are used in this report.

3 Answer groups and compact evidence

Let $x_{qi} \in \mathbb{R}^{1024}$ denote the supplied candidate embedding. Candidates with the same valid parsed answer form G_{qg}, with answer a_{qg}. An unparseable answer belongs to no valid group. Equivalent forms such as 0.5 and 1/2 are pooled exactly, rather than by approximate string or embedding similarity. The semantic representation of group g is

$$\bar{x}_{qg} = \frac{1}{|G_{qg}|} \sum_{i \in G_{qg}} x_{qi}.$$

This aligns the action space with the scorer: all members of one numeric group have the same final-answer correctness. However, averaging can discard useful trace differences. Pooling is consequently a hypothesis to test, not a presumed improvement over candidate-level scoring.

Table 2: Nine descriptors appended to the group mean; no learned text encoder.

Descriptor	Definition or implementation
Support fraction	$\lvert G_{qg}\rvert/4$, with the original four slots as denominator.
Log variance	$\log(1+v)$, $v = \frac{1}{1024\lvert G\rvert} \sum_{i \in G} \lVert x_{qi} - \bar{x}_{qg}\rVert^2$.
Pairwise cosine	Mean cosine over distinct within-group candidate pairs; zero for a singleton. Norms are floored at 10^{-12}.
Signed log magnitude	$\ell(a) = \operatorname{sign}(a)\log(1+\lvert a\rvert)$.
Integer flag	Whether the canonical rational answer has denominator one.
Zero flag	Whether the answer equals zero.
Negative flag	Whether the answer is negative.
Exact question number	Whether the answer occurs among rational numeric tokens extracted from the question.
Relative magnitude	$\ell(a/M_q)$, where M_q is the largest absolute extracted question number; zero if $M_q = 0$.

The final feature vector $f_{qg} = [\bar{x}_{qg}; h_{qg}]$ has 1033 dimensions. Statistics such as group support and variance describe agreement; numeric descriptors provide low-cost information about the form and scale of an answer. They are *soft learned evidence*, not hard rules. An integer answer is not automatically correct, and a minority answer is not automatically wrong.

Question numbers are extracted with a bounded regular expression and exact rational conversion (at most 20,000 characters and 512 tokens). This does not interpret units, percentages or the role of a quantity. For example, detecting an answer already present in a question is different from proving that it is the requested quantity. These restrictions are both engineering choices and limits of the evidence available to the model.

The final model contains *no local equation checker* and no explicit question-mean-centering branch. In the linear embedding component, a common question-level offset cancels in pairwise differences; the appended cosine descriptor is not translation-invariant. Earlier nonlinear models used explicit centering; that earlier design should not be attributed to the final ranker. Likewise, the nine-dimensional block mixes group statistics and answer form, so its measured benefit cannot be credited to answer form alone.

4 Preference objective and implementation

For informative question q, there is exactly one correct numeric answer group g_q^+ and $K_q - 1$ incorrect valid groups. A linear score is $s_{qg} = w^\top z_{qg}$, where z is standardized f. The Bradley–Terry model uses $\sigma(s_{qg^+} - s_{qg})$ for a pairwise preference [3]. We minimize

$$\mathcal{L}(w) = \frac{1}{|H|} \sum_{q \in H} \frac{1}{K_q - 1} \sum_{g \neq g_q^+} \log\Big(1 + \exp\{-w^\top (z_{qg_q^+} - z_{qg})\}\Big) + \frac{\lambda}{2}\|w\|_2^2, \qquad \lambda = 0.01. \tag{1}$$

Each question has total pair weight one; its wrong answer groups share that weight uniformly. A question with three wrong answers therefore does not count three times as much as a question with one. Multiple copies of a wrong numeric answer are collapsed rather than receiving extra negative weight.

For the candidate-level control, all correct/incorrect candidate pairs are used, but a pair with incorrect group g has weight $1/[(K_q - 1)|G_q^+||G_{qg}|]$. This preserves equal question and wrong-answer-group mass despite repeated positive or negative candidates. It makes the main representation comparison more meaningful than an unweighted pair expansion.

Train-only normalization. The 1024 embedding means and scales are estimated from all four original candidates of informative training questions, including embeddings of invalid-answer candidates. This convention is shared by candidate and group controls. The nine appended statistics are normalized using valid answer groups in the same training support. Standard deviations are floored at 10^{-6}. Pooling commutes with the affine embedding transform; the different descriptor block has its own statistics. Validation and held-out folds never supply these statistics.

Table 3: The original full-training final fit. Times exclude data loading.

Informative questions / original candidate embeddings	2114 / 8456
Valid groups / preference pairs	5172 / 3058
Total pair weight / coefficients	2114 / 1033
L-BFGS-B iterations / function evaluations	71 / 80
Maximum absolute final gradient	5.90×10^{-7}
Recorded fit time on the local CPU	1.0943 s

Optimization uses deterministic zero initialization, analytic gradients and SciPy L-BFGS-B: at most 500 iterations, `ftol=1e-10`, `gtol=1e-7`, and `maxls=50`. There is no random training seed, intercept, early stopping or validation fitting. Numerically stable softplus avoids exponential overflow.

At inference, we select the highest-scoring valid group and return its first member. Groups and exact ties use first-occurrence order. Thus the selected *numeric answer* is insensitive to duplicate-member identity, but exact ties can depend on order. On an all-invalid input, a defined embedding-only fallback chooses a candidate. The method accepts arbitrary input bank sizes and reads only passed texts/embeddings and the frozen bundle.

5 Main results and controlled interpretation

The main comparison holds informative support, folds, normalization convention, L2 and question/pair weighting fixed. Each representation has three fold fits and one full fit; validation follows freezing. The difference between the last two rows is the added descriptor block, not a change of optimizer or support.

Table 4: Matched linear comparisons. OOF and validation have different supports.

Representation	OOF /2114	OOF %	Val /355	Val %
Candidate embeddings	1682	79.56	288	81.13
Answer-group mean	1596	75.50	283	79.72
Group mean + nine features	**1713**	**81.03**	**299**	**84.23**

Pure pooling loses 86 OOF questions and five validation questions relative to candidate preferences. Group size and heterogeneity matter: a mean alone cannot describe whether it summarizes one trace or several disagreeing-quality traces. Adding the block recovers and exceeds the candidate control, gaining 117 OOF questions over pure pooling (232 fixes, 115 regressions). The fold nets are $+40, +31, +46$; the validation net is $+16$. Against candidate preferences, the nets are $+31$ OOF ($+6, +13, +12$ by fold) and $+11$ validation. These are consistent directions in this protocol, not an independent multi-dataset study.

Table 5: Historical validation context. These are not fully matched causal controls.

Method	Full val /798	Informative val /355
Initial numeric majority vote	688	245
Independent shared MLP (E1)	717	281
Question-centered MLP, raw selection (E2)	727	288
Same frozen E2, invalid answers excluded	733	290
Grouped nonlinear head (E3, seed 42)	734	291
Fixed E2/E3 equal fusion (seed 17)	741	298
Final group preference model	**742**	**299**

The initial majority result establishes that learned selection helps on this bank. The raw E2-to-E3 increase mixes grouping with exclusion of unparseable answers: matching the latter leaves only one validation question. The final model is also only one validation question above the historical best fusion. Neither observation justifies a strong architecture-only superiority claim. Older MLPs use different support/statistic conventions and selected epochs.

Other negative findings. On the same compact 48-dimensional feature set, linear classification answered 1652 OOF questions correctly; CatBoost classification obtained 1623/1627 across two seeds and weighted pairwise trees 1627/1638. Ranking slightly helped the trees, but did not close the gap to that linear control. Set-interaction networks, compared with matched independent networks, changed OOF correctness by $+3/-14$ and validation by $-3/+6$ across two seeds. These failures discouraged added complexity; they do not establish that every tree or set architecture must fail.

6 Feature ablations and bounded refinements

Each ablation deletes one of the nine columns and refits on the original folds, with L2 fixed at 0.01. The column is zeroed in standardized training differences before optimization; it is not merely removed from a previously trained model. Other normalization statistics and the frozen feature schema remain unchanged.

Table 6: Drop-one ablations relative to 1713/2114. Fold entries are net changes.

Removed feature	OOF correct	Net	Fold 1	Fold 2	Fold 3
Support fraction	1716	+3	+2	−2	+3
Log variance	1702	−11	−3	−5	−3
Mean pairwise cosine	1713	0	+6	−1	−5
Signed log magnitude	1713	0	+2	0	−2
Integer flag	1696	−17	−7	−9	−1
Zero flag	1707	−6	+1	−2	−5
Negative flag	1710	−3	0	−1	−2
Exact question number	1713	0	0	0	0
Relative magnitude	1705	−8	0	−4	−4

Integer status and within-group variance have the largest measured conditional effects, with negative deletion nets in all three folds. The exact-question-number feature has no net effect, but this does not show that its predictions are identical or that the feature is universally useless. Correlated evidence can be redundant. Drop-one effects measure utility conditional on the other eight features and L2, not independent causal effects of each property.

Deleting support fraction is the best observed deletion: 12 fixes and nine regressions, net +3. It leaves validation at 299/355, tied with the full block. It falls short of the predeclared +11 OOF adoption threshold, so all nine features remain. The threshold is an engineering guard against repeated small changes; it is not a significance test or a correction for all past experiments.

Table 7: A bounded representation/L2 probe; original final is retained.

Representation	L2	OOF /2114	Net vs. final
Group + nine (original)	0.01	1713	0
Candidate embedding + same group descriptors	0.01	1708	−5
Group + nine	0.003	1709	−4
Group + nine	0.03	1723	+10

The same +11 gate was not met by the stronger-regularization probe. We kept the original L2=0.01 configuration before its official test. This does not prove 0.01 is the globally optimal parameter; it documents a bounded decision rule and avoids claiming a small reused-OOF gain as decisive evidence. No additional representation, ensemble, arithmetic correction or deletion was added to the final checkpoint.

7 Reliability, selection bias and failure boundaries

OOF predictions do not make subsequent rule selection unbiased. We considered replacing low-margin group decisions with candidate-level BT or unique-majority selection. The margin is the raw logit gap between the best two valid answer groups. Threshold candidates were fixed in advance, including turning fallback off. A one-group input never needs fallback; majority ties retain the anchor choice.

Table 8: The same fallback can look helpful when selected and evaluated on all OOF.

Rule	All-OOF selected /2114	Nested OOF /2114	Val /355
No fallback	1713	1713	299
Candidate-BT fallback	1728	1711	297
Majority fallback	1716	1709	297

Nested evaluation adds three inner folds inside each outer training partition. Both base models, their statistics and the threshold are fitted without that outer held-out partition. Candidate fallback then changes by $-1, +2, -3$ across outer folds (28 fixes and 30 regressions overall); majority fallback changes by $0, -4, 0$. The +15 all-OOF candidate gain is therefore not a repeatable estimate of an independent gain. We reject both rules. Final full-OOF thresholds were only training selections; validation did not rescue either route. This example is more informative than reporting only the best threshold score.

Interpreting the validation–test gap. Full validation is 92.98%, whereas informative validation is 84.23% and official test is 82.87%. The comparable difference is approximately 1.36 percentage points, not the full 10.11-point difference. Much of the apparent gap is explained by the easy-question mixture. This does not rule out remaining distribution shift, finite-bank variation, validation adaptation or shared model biases.

Nominal Wilson 95% intervals are approximately 80.07–87.65% for 299/355 and 79.19–86.02% for 387/467. They describe binomial sampling scale under a simple independent-question assumption; they do not account for adaptive development, public-bank reuse or dataset dependence. They are not a guarantee of future accuracy and cannot turn this public test into a blinded experiment. Likewise, two seeds or two fold assignments over the same questions are not independent new datasets.

What the model cannot certify. Its scores are relative preferences, not calibrated probabilities that an answer is correct. Similar traces may share an incorrect interpretation; exact numeric agreement can reproduce the same mistake. The number features do not identify the requested unit or quantity, and integer/magnitude priors can fail under a new problem distribution. All these banks contain a correct candidate, so there is no missing-answer ceiling here. A correct minority in a 3–1 split is also selectable. The current errors do not establish an impossibility bound: reaching 85% on this test would need 397 correct, ten more than observed.

8 Resources, reproduction and conclusions

The frozen bundle contains `config.json` (79 bytes) and `weights.npz` (26,110 bytes), including all means, scales and coefficients. Its total, 26,189 bytes, is about 0.025 MiB versus the course's 16 MiB limit. The stored official test measurement is 1.3019 s from fresh prediction-process startup through independent scorer exit, including imports, data/assets and preprocessing. The machine was an Intel Core i7-12700H on Windows 11, with four threads set across numerical libraries. This is local evidence, not certification on the reference Xeon Platinum 8457C or a replacement for its 30 s eligibility test.

Table 9: Selected recorded development costs, not total project wall-clock time.

Stage	New base fits	Recorded fit seconds
Matched linear/set comparison (E17)	28	32.2933
Nested fallback validation (E20)	18	16.8306
Drop-one feature study plus one full fit (E21)	28	42.5122

These are selected fit costs, excluding implementation, review and some orchestration. They are not complete lifetime search cost. Training needs Python 3.12, NumPy 2.5.3 and SciPy 1.18.1; inference needs only NumPy and supplied tools. The final optimizer is deterministic; comparison folds use seed 20261028.

Reproduction. The public repository contains a clean final-only entry, the checkpoint, compact numerical records, tests and this PDF/source. Course data is installed from the official ZIP and checked against its unchanged manifest. From an activated environment, use:

```
python -m pip install -r requirements-training.txt
python code/fetch_data.py
python code/train.py --config artifacts/final/config.json --run-dir runs/retrain
python code/train.py --config artifacts/final/config.json --run-dir runs/cv --reproduce-comparisons
```

The fixed comparison mode repeats the three main representations and nine drop-one ablations on the original fold algorithm. Official prediction and independent scoring commands, thread settings, asset hashes and the frozen version are in the README/metadata. Evaluation of the supplied checkpoint requires no retraining, old run folders, cached predictions or network access. Platform-specific numerical differences are possible; the supplied checkpoint is the artifact associated with the recorded official score.

Conclusion. The supported improvement is the descriptor block combined with group preferences. Compare the same disagreement population, normalize pair weights, fit statistics inside folds, and evaluate rule selection on additional held-out data. These findings do not certify mathematical reasoning.

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References

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